

0. 考试导航 Exam Map

看到题目	立刻做什么
Draw BN / joint	一个节点一个因子； root 写 prior，有父节点写 conditional
How many params	Boolean node with m parents: 2^m params; full joint: $2^n - 1$
$P(Q \mid e)$	分 query/evidence/hidden; 写 $\alpha \sum_H \prod_i P(X_i \mid Pa_i)$
Given CPT numbers	取对应行; 要 false 用 $1 - P(true)$; 最后乘或 normalize
Factor operation	Join= 对齐变量相乘; Eliminate=sum out 让变量消失
Variable elimination	选 hidden 变量; join 含它的 factors; sum out; 放回; normalize
Order comparison	看最大 intermediate factor; 变量越多越贵, k^n grows fast

1. BN Joint Distribution

中文主线: BN 用 DAG + CPT 表示完整联合分布，不直接存 full joint table。
A Bayesian Network represents a joint distribution as a product of local conditional distributions.

$$P(x_1, \dots, x_n) = \prod_i P(x_i \mid Parents(X_i))$$

写 joint 模板: 对每个 node 写一个因子; root node 写 $P(X)$; 有父节点写 $P(X \mid Pa(X))$ 。

```
| A -> D <- B,    B -> E <- C
| P(A, B, C, D, E)
| =P(A)P(B)P(C)P(D|A,B)P(E|B,C)
```

参数数目 params: Boolean variable: 每个 parent assignment 只需 1 个数 $P(X = T \mid parents)$, 因为 $P(F) = 1 - P(T)$ 。若有 m 个 Boolean parents: 2^m params。无独立性 full joint: $2^n - 1$ params。

English sentence: *The joint distribution factorizes according to the BN structure. Conditional independences reduce the number of CPT parameters.*

2. Inference 是在算什么

Inference: 根据 evidence 计算 query 的 posterior probability。
Inference calculates useful probabilities, especially posterior probabilities, from a fixed BN.

三类变量: 对 $P(Q \mid E = e)$:

Query	竖线左边	Q
Evidence	竖线右边已知	$E = e$
Hidden	All - Query - Evidence	要被求和消去

后验展开 posterior:

$$P(Q \mid e) = \frac{P(Q, e)}{P(e)} = \alpha P(Q, e)$$
$$P(Q \mid e) = \alpha \sum_{h_1} \cdots \sum_{h_r} \prod_i P(x_i \mid Pa_i)$$

考场英文: *Let Q be the query, e the evidence, and H the hidden variables. Then $P(Q \mid e) = \alpha \sum_H \prod_i P(x_i \mid Parents(X_i))$, where α normalizes the result.*

3. Enumeration 做题步骤

Enumeration: 枚举 hidden variables 所有取值，对 joint 求和，再 normalize。
Enumeration sums over all assignments of hidden variables and then normalizes.

- 写 BN joint factorization。
- 标出 Query / Evidence / Hidden。
- 写 $P(Q, e) = \sum_H P(Q, H, e)$ 。
- 用 BN joint 代入 CPT product。
- 对 query 的每个值算一个未归一化数; divide by total。

Alarm 模板: $P(B \mid +j, +m)$

```
| P(B,E,A,J,M)=P(B)P(E)P(A|B,E)P(J|A)P(M|A)
| Query: B
| Evidence: J=+j, M=+m
| Hidden: E,A
| P(B|+j,+m)=alpha SUM_e SUM_a
| P(B)P(e)P(a|B,e)P(+j|a)P(+m|a)
```

坑: 有 evidence 时，不要 sum evidence; 只固定它的值。Hidden 才 sum out。

4. Factor 是什么

Factor: 关于若干变量的中间表，不一定 normalized。
A factor is a multi-dimensional table over a set of variables. It need not be a valid probability distribution.

维度判断: 大写未固定变量个数 = factor 维度。 $P(W \mid T)$ has variables W, T (2D); $P(W \mid cold)$ only W (1D)。

Join factors: 对共同变量对齐，逐项相乘；新 factor 的变量集是 union。

$$f_3(R, T) = f_1(R) f_2(T, R)$$

Eliminate / sum out: 对某变量所有取值求和，让它从 factor 消失。

$$g(T) = \sum_r f(r, T)$$

```
| Boolean sum-out:
| g(T)=f(R=0,T)+f(R=1,T)
```

4A. 手算补充: Factor / Evidence / Normalize

Join: 输出变量 = 两个 factor 变量的 union；每行按相同变量对齐后相乘。

```
| f1(A): A0=.8,A1=.2; f2(A,B): (A0,B0)=.9,(A0,B1)=.6,(A1,B0)=.5,(A1,B1)=.1
| join f(A,B): .72,.48,.10,.02
```

Sum out: 消 A 就把相同剩余变量的 $A = 0, A = 1$ 两行相加: $g(B0) = .72 + .10 = .82$, $g(B1) = .48 + .02 = .50$ 。**Join 变宽; sum-out 变窄。**

Evidence instantiate: Evidence 固定值，不求和。例: factor $P(D \mid A, B)$ 且 $B = T$, 只保留 $B = T$ 的行，变成 $P(D \mid A, B = T)$ ，可看成 over D, A 的 factor。

Normalize: 未归一化向量 $u = [u_F, u_T]$:

$$P(F \mid e) = \frac{u_F}{u_F + u_T}, \quad P(T \mid e) = \frac{u_T}{u_F + u_T}, \quad \alpha = 1/(u_F + u_T)$$

例: $u_F = .0576, u_T = .0024 \Rightarrow P(F \mid e) = .96$ 。

Count sum-products: 消 H 后的新 factor 含 r 个 Boolean 变量 $\Rightarrow 2^r$ 行，也就是 2^r 次 sum-product。
 $P(X_3 \mid X_2)$ 有 2 个变量 $\Rightarrow 4$ 次; $P(X_4 \mid X_1, X_2)$ 有 3 个变量 $\Rightarrow 8$ 次。

5. Variable Elimination 标准算法

VE 核心: 不先造完整 joint; 边 join 边 eliminate，尽早消 hidden。
Variable Elimination interleaves joining and marginalization to keep intermediate factors smaller.

```
| Variable-Elimination(0,e,BN):
|   factors = local CPTs instantiated by evidence e
|   for H in elimination order:
|     relevant = all factors containing H
|     joined = Join(relevant)
|     new = SumOut(joined, H)
|     remove relevant; add new
|   result = Join(all remaining factors)
|   return Normalize(result)
```

Evidence 怎么处理: 先在所有包含 evidence 的 factors 里固定值 (select rows / reduce dimension)。只 eliminate 非 query、非 evidence 的变量。

为什么比 enumeration 快: *Enumeration first joins the whole joint distribution and then sums out hidden variables. VE joins only the factors involving the current hidden variable and immediately sums it out. VE is still exponential in the worst case, but often creates much smaller intermediate factors.*

6. Elimination Order 判断

复杂度看最大中间 factor: 若 domain size 为 k , factor 含 r 个变量，表大约 k^r 。Join 后变量越多越贵。
Good order 先消叶子/局部变量，largest factor small
Bad order 过早消连接多的变量，creates large factors
结论 *The largest intermediate factor dominates the time and space cost.*

一句话答案: 好顺序会让 intermediate factors 变量少；坏顺序会产生含很多变量的大表。Because factor size is exponential in the number of variables it contains.

7. 小链例子: $R \rightarrow T \rightarrow L$

Joint: $P(R, T, L) = P(R)P(T \mid R)P(L \mid T)$

Naive enumeration: $P(L) = \sum_r \sum_t P(r)P(t \mid r)P(L \mid t)$

VE: 先消 R: $g(T) = \sum_r P(r)P(T \mid r)$
 $P(L) = \sum_t g(t)P(L \mid t)$

差别: Enumeration 倾向于先合成 R, T, L 大表; VE 先把 R 变成关于 T 的小表。

8. 数字 CPT 题: 万能流程

给 CPT 算概率时:

- 写 joint product，按图取因子。
- 对每个变量取题目指定值；若问 F 且表给 $P(T)$ ，用 $1 - P(T)$ 。
- 全部已知的 joint: 直接乘。
- 条件概率 $P(A = a \mid e)$: 算每个 A 取值的 joint，再 normalize。

Conditional normalization 模板

$$P(A = F \mid e) = \frac{P(A = F, e)}{P(A = F, e) + P(A = T, e)}$$

English: *Compute the unnormalized joint probability for each possible value of the query variable, then normalize by dividing by their sum.*

9. 常见坑 Top Checks

- False probability: $P(X = F) = 1 - P(X = T)$ 。
- $P(D = T \mid A, B)$ 表中 A, B 顺序要对准题目。
- Evidence 固定，不参与 summation。
- Hidden = 全部变量 - query - evidence。
- VE 每次只 join containing current H 的 factors。
- Sum out 后 factor 不再含 H 。
- 最后得到的是未归一化 posterior 时，必须 normalize。

10. Tutorial A: Medical BN

题型: Draw BN / write joint / count parameters / explain observations.
Diseases D_1, D_2, D_3 independent. Symptoms: S_1 by D_1 ; S_2 by D_1, D_2 ; S_3 by D_1, D_3 ; S_4 by D_3 .

```
| D1 -> S1
| D1 -> S2 <- D2
| D1 -> S3 <- D3
| D3 -> S4
```

Joint expression

```
| P(D1,D2,D3,S1,S2,S3,S4)
|=P(D1)P(D2)P(D3)
| P(S1|D1)P(S2|D1,D2)
| P(S3|D1,D3)P(S4|D3)
```

Parameter count

CPT	params	why
$P(D_1), P(D_2), P(D_3)$	1 + 1 + 1	no parent
$P(S_1 D_1)$	2	1 parent
$P(S_2 D_1, D_2)$	4	2 parents
$P(S_3 D_1, D_3)$	4	2 parents
$P(S_4 D_3)$	2	1 parent
Total	15	

No conditional independences: clique CPT params 1 + 2 + 4 + 8 + 16 + 32 + 64 = 127, same as full joint $2^7 - 1$.

Observing $S_4 = T$: only learns about D_3 . If also know $S_3 = T$, then $S_4 = T$ also gives information about D_1 via explaining away at S_3 ; still not D_2 .

11. Tutorial B: BN with A,B,C,D,E

Structure:

```
| A -> D <- B
| B -> E <- C
| P(A,B,C,D,E)=P(A)P(B)P(C)P(D|A,B)P(E|B,C)
```

Priors: $P(A = T) = 0.2, P(B = T) = 0.5, P(C = T) = 0.8$.

CPTs given in tutorial

A	B	$P(D = T A, B)$	B	C	$P(E = T B, C)$
F	F	0.9	F	F	0.2
F	T	0.6	F	T	0.4
T	F	0.5	T	F	0.8
T	T	0.1	T	T	0.3

Q1: all five variables false

```
| P(F,F,F,F,F)
|=P(AF)P(BF)P(CF)P(DF|AF,BF)P(EF|BF,CF)
|= (0.8) (0.5) (0.2) (0.1) (0.8)=0.0064
```

$P(D = F | F, F) = 1 - 0.9 = 0.1$; $P(E = F | F, F) = 1 - 0.2 = 0.8$.

Q2: $P(A = F | B = T, C = T, D = T, E = T)$

No hidden variables; enumerate the two values of A.

```
| uF=P(AF,BT,CT,DT,ET)
| =(0.8) (0.5) (0.8) (0.6) (0.3)=0.0576
| uT=P(AT,BT,CT,DT,ET)
| =(0.2) (0.5) (0.8) (0.1) (0.3)=0.0024
| P(AF|BT,CT,DT,ET)=uF/(uF+uT)
| =0.0576/0.0600=0.96
```

PDF typo: final query should be $A = F$, answer 0.96.

12. Tutorial C: VE Network

Structure:

```
| X1 -> X3 <- X2
| X3 -> X4
| P(x1,x2,x3,x4)=P(x1)P(x2)P(x3|x1,x2)P(x4|x3)
| Goal: find P(x4), all Xi in {0,1}
```

Priors: $P(X_1 = 0) = 0.3, P(X_1 = 1) = 0.7; P(X_2 = 0) = 0.6, P(X_2 = 1) = 0.4$.

Naive marginalization

$$P(x_4) = \sum_{x_1} \sum_{x_2} \sum_{x_3} P(x_1) P(x_2) P(x_3 | x_1, x_2) P(x_4 | x_3)$$

For each x_4 , sum over $k^3 = 8$ assignments. General cost grows exponentially.

13. VE Order 1: Eliminate X_1 first

Result network: $X_2 \rightarrow X_3 \rightarrow X_4$.

$$P(x_3 | x_2) = \sum_{x_1} P(x_3 | x_1, x_2) P(x_1)$$

Example: $P(X_3 = 0 | X_2 = 0) = 0.5(0.3) + 0.9(0.7) = 0.78$

Resulting CPT

x_3	x_2	$P(x_3 x_2)$
0	0	0.78
0	1	0.41
1	0	0.22
1	1	0.59

Continue to $P(x_4)$ if needed

$$\begin{aligned} P(x_3) &= \sum_{x_2} P(x_2) P(x_3 | x_2) \\ P(X_3 = 0) &= 0.6(0.78) + 0.4(0.41) = 0.632 \\ P(X_3 = 1) &= 0.368 \\ P(X_4 = 0) &= 0.7(0.632) + 0.1(0.368) = 0.4792 \\ P(X_4 = 1) &= 0.5208 \end{aligned}$$

Sum-product counts: $p(x_3 | x_2)$: 4; $p(x_3)$: 2; $p(x_4)$: 2. Total 8.

14. VE Order 2: Eliminate X_3 first

Result network: $X_1 \rightarrow X_4 \leftarrow X_2$.

$$P(x_4 | x_1, x_2) = \sum_{x_3} P(x_4 | x_3) P(x_3 | x_1, x_2)$$

Example: $P(X_4 = 0 | X_1 = 0, X_2 = 0) = 0.7(0.5) + 0.1(0.5) = 0.40$

Resulting CPT

x_4	x_1	x_2	$P(x_4 x_1, x_2)$
0	0	0	0.40
0	0	1	0.22
0	1	0	0.64
0	1	1	0.40
1	0	0	0.60
1	0	1	0.78
1	1	0	0.36
1	1	1	0.60

Sum-product counts: $p(x_4 | x_1, x_2)$: 8; $p(x_4 | x_1)$: 4; $p(x_4)$: 2. Total 14. Therefore order 1 is preferable.

15. CPT Values Used in Tutorial C

x_3	x_1	x_2	$P(x_3 x_1, x_2)$	x_4	x_3	$P(x_4 x_3)$
0	0	0	0.5	0	0	0.7
0	0	1	0.2	0	1	0.1
0	1	0	0.9	1	0	0.3
0	1	1	0.5	1	1	0.9
1	0	0	0.5			
1	0	1	0.8			
1	1	0	0.1			
1	1	1	0.5			

How to compute any row: fix the variables in the row, multiply the relevant CPT entries, and sum over the eliminated variable. Each output row is one sum-product calculation.

16. 答题模板: Posterior / Enumeration

Question: compute $P(Q | \text{evidence } e)$

- Factorization:
 $P(\text{all}) = \text{PRODUCT}_i P(X_i | \text{Parents}(X_i))$
- Classify variables:
Query = ...
Evidence = ...
Hidden = all - query - evidence
- Expand:
 $P(Q|e) = \alpha \sum_{\text{hidden}} \text{PRODUCT}_i P(X_i | \text{Pa}_i)$
- If numeric:
compute one unnormalized number for each Q value
normalize: $\text{ans}(q) = u_q / \sum_q u_q$

17. 答题模板: Variable Elimination

```
| Initial factors:
| f1=..., f2=..., ...
| Evidence:
|   instantiate/select rows if any
| Elimination order:
|   H1, H2, ...
|
| For H1:
|   join factors containing H1:
|   g(...,H1,...)=...
|   sum out H1:
|   h(...)=SUM_H1 g(...,H1,...)
|   replace old factors by h
|
| Repeat. Join remaining factors. Normalize if conditional.
```

写解释: The first order is better because its largest intermediate factor contains fewer variables, so it requires fewer sum-product calculations.

18. 英文金句 Directly Copy

Inference: Inference computes posterior probabilities of query variables given observed evidence.
Enumeration: Inference by enumeration sums over all possible assignments of hidden variables and normalizes the result.
Factor: A factor is a table over a set of variables and does not necessarily represent a normalized probability distribution.
Join: Joining factors multiplies them pointwise over the union of their variables.
Eliminate: Eliminating a variable sums over all its possible values, removing it from the factor.
VE: Variable elimination avoids constructing the full joint distribution by joining relevant factors and summing out hidden variables early.
Order: The computational cost is dominated by the largest intermediate factor; therefore the elimination order matters.

19. 考场最后检查

- Joint 是否每个节点正好出现一次?
- Root prior / child conditional 是否写反?
- 给 F 的概率是否用了 complement?
- Conditional 是否对 query 所有取值 normalize?
- Summation 是否只 over hidden, 不 over evidence?
- VE 是否每步删除 old factors, 再放回 new factor?
- Order comparison 是否写了具体 counts 或 largest factor?
- 表格 row 的 parent assignment 是否对准?

20. 30 秒速记

BN: 一个节点一个 CPT factor。
Posterior: $P(Q|e) = \alpha \sum_H \prod_i P(X_i | \text{Pa}_i)$ 。
Enumeration: 全部 hidden 组合都加一遍。
Factor: table, may be unnormalized.
Join: multiply. Eliminate: sum out.
VE: join relevant factors, sum out immediately.
Order: largest factor decides cost.
Multi-query: query variables all stay; normalize over their joint assignments.
Non-Boolean: replace 2^n by product of domain sizes.